

Personality development through natural language

Kevin Lanning¹✉, Rachel E. Pauletti², Laura A. King³ and Dan P. McAdams⁴

Ego level is a broad construct that summarizes individual differences in personality development¹. We examine ego level as it is represented in natural language, using a composite sample of four datasets comprising nearly 44,000 responses. We find support for a developmental sequence in the structure of correlations between ego levels, in analyses of Linguistic Inquiry and Word Count (LIWC) categories² and in an examination of the individual words that are characteristic of each level. The LIWC analyses reveal increasing complexity and, to some extent, increasing breadth of perspective with higher levels of development. The characteristic language of each ego level suggests, for example, a shift from consummatory to appetitive desires at the lowest stages, a dawning of doubt at the Self-aware stage, the centrality of achievement motivation at the Conscientious stage, an increase in mutuality and intellectual growth at the Individualistic stage and some renegotiation of life goals and reflection on identity at the highest levels of development. Continuing empirical analysis of ego level and language will provide a deeper understanding of ego development, its relationship with other models of personality and individual differences, and its utility in characterizing people, texts and the cultural contexts that produce them.

The notion that personality remained unchanged through adulthood was once widely accepted³. More recently, longitudinal investigations of traits have revealed both normative changes with development (maturation) and individual differences in these trends⁴. All told, the trait approach has fostered a large and important literature on some of the ways in which adult personality changes over time^{5–8}. Yet, traits represent only one aspect of adult personality. Conceptualized as relatively enduring characteristics, traits are, by definition, less malleable than other features of the individual. Consider, for instance, an extravert who has experienced a potentially transformative life event, such as an unexpected divorce or the loss of a child. The person may emerge from that experience still an extravert but different in ways that extraversion does not capture. With difficult experience might come a change in the person's stance towards the self and the world—perhaps wiser, more compassionate and more likely to view life choices in relativistic terms, with insight into the vulnerability implied in human existence. Such a person has experienced an essential change that is not necessarily relevant to traits, but that reflects engagement with challenging experiences. How can such changes be incorporated into our growing understanding of personality development?

Loevinger¹ introduced the concept of ego development to capture personality development as a process from a relatively simple to an increasingly differentiated and complex frame of reference in relation to oneself and the world. At lower levels of ego development,

the person's perspective is simple, dominated by impulse or controlled by external factors. As ego development increases, the self and the world are seen as increasingly complex. This view includes the capacity for contextualizing life's 'right answers' and a deeper sense of compassion towards others, often grounded in difficult life experiences⁹. For Loevinger, the ego is "the striving to master, to integrate, and make sense of experience" (see p. 59 in ref. ¹).

Ego development has traditionally been measured using the Washington University Sentence Completion Test (WUSCT)¹⁰. On the WUSCT, participants provide responses to open-ended verbal stems such as "A good father...", and trained raters subsequently code the responses for ego level. (For this stem, the response "...is a good man" would be coded at the low Self-protective stage, "...truly cares and loves his children" at the intermediate Self-aware stage and "...should respect children and wife's individuality" at the high Autonomous stage; see Table 1).

Research using the WUSCT supports the notion that ego level increases with age^{11,12}, is predicted by actively coping with difficult times^{13–15} and, in narrative accounts, is manifest in reflective stories that suggest actively struggling to make sense of experience¹⁶. In addition, such narrative reflection predicts prospective increases in ego development⁹. For example, in a sample of stories written by parents about their experience of finding out their child had Down Syndrome, narratives revealing a paradigmatic shift in philosophy of life were associated with concurrent ego level; moreover, this narrative feature was associated, prospectively, with increasing ego level over 2 years¹³. Ego development is correlated positively with self-reports of openness to experience¹⁶, ego strengths¹⁷, intuition¹⁸, identity achievement^{19,20} and more ethical decision making²¹, supporting its construct validity²². However, none of these relations is so high as to suggest that an established self-report measure of these other characteristics might serve as a proxy for the WUSCT. Furthermore, ego level is unrelated to self-reports of stress-related growth¹³.

For all the richness and utility of Loevinger's theory, the number of empirical studies of the construct is in fact relatively modest, in large part because the WUSCT is time consuming and labour intensive for both the respondent and the assessor. The process of scoring includes both a training phase and, subsequently, frequent references to a lengthy manual. Surely, the level of detail in that manual, comprising >7,500 responses, has led the WUSCT to have substantial inter-rater and inter-item reliability²². However, using the WUSCT is onerous. One goal of the present investigation is to explore whether recent developments in the analysis of language might be leveraged to facilitate the measurement of this rich construct.

Although assessing personality and psychological constructs using language is not new^{23–25}, recent advances in computational methods,

¹Florida Atlantic University, Jupiter, FL, USA. ²Lynn University, Boca Raton, FL, USA. ³University of Missouri, Columbia, MO, USA. ⁴Northwestern University, Evanston, IL, USA. *e-mail: klanning@fau.edu

Table 1 | Covarying cognitive, social and moral facets in the stages of ego development

Stage or level	Cognitive facet (concern)	Social facet (breadth of perspective)	Moral facet (impulse control, constraint)
Impulsive (2)	Dichotomous experience (good versus bad)	Egocentrism, solipsism	Urges, gratification
Self-protective (3)	Trouble, control	Manipulation, wariness	Opportunism, exploitativeness
Conformist (4)	Appearances, convention	Cooperation, loyalty	Obedience, respect for authority
Self-aware (5)	Problems, adjustment	Helpfulness, self-awareness	Contingencies, exceptions to rules
Conscientious (6)	Traits, achievement	Responsibility, concern	Self-criticism, conscience
Individualistic (7)	Development, roles	Mutuality, emotional dependence	Tolerance, differentiation
Autonomous/Integrated (8/9)	Fulfillment, identity	Interdependence, individuality	Coping with conflict, complexity

Table adapted from ref. ¹⁰.

the availability of large datasets and increasingly frequent interdisciplinary collaborations among social and computational scientists have led to dramatic improvements in the analysis of this form of behaviour²⁶. One of the most important contemporary tools for examining large bodies of textual data is the Linguistic Inquiry and Word Count (LIWC), which breaks down texts based on categories that include both syntactical composition and psychological meaning². In the recently published fourth revision of the LIWC, approximately 6,400 terms are classified into 81 categories ranging from first-person singular pronouns (I, me) to drives such as power²⁷ (superior, bully).

A distinction may be drawn between the closed-language, dictionary-based approach of the LIWC and an open approach in which the frequencies of words are compared for, or correlated with, characteristics of interest²⁸. This approach has been applied to texts ranging from political speeches and religious sermons to the online flow of Facebook status updates and Twitter posts, and has revealed insights concerning personality characteristics, such as the five factors^{28,29}, psychopathy³⁰ and mood³¹, as well as external criteria ranging from coronary heart disease to differences among regions of the United States³².

Because it is a rich and implicative construct that is assessed via samples of language (sentence completions), ego development is a particularly suitable candidate for the quantitative study of personality via text analysis. This attempt to bridge the areas of personality and development using contemporary text analysis is not without precedent: for example, one study of the relationship between LIWC categories and the content of Adult Attachment Interviews revealed that anger and inhibition were both characteristic of individuals classified as 'preoccupied'³³. As with the Adult Attachment Interviews, a response to the WUSCT is a self-expression, a speech act whose intention is, in part, to articulate a facet of identity. Sentence completions arguably convey characteristics which are more fundamentally endogenous and idiographic than the simple endorsement of questionnaire statements^{34,35}.

In this project, we explicate the construct of ego development using descriptive and exploratory analyses of a large composite sample of responses to the WUSCT. We consider LIWC categories and individual words, examining their relationships with ego maturation and the characteristics of individual ego levels. More specifically, our analyses include five separate sections: (1) an examination of the claim that ego level constitutes an ordered sequence, (2) an assessment of relationships between ego level and LIWC categories related to cognitive complexity and (3) a similar assessment of relationships between ego level and breadth of perspective, (4) an exploration of linear and non-linear relationships between additional LIWC categories and ego level and (5) an analysis of the characteristic language associated with each ego stage or level.

(1) The primary evidence for ego level as a developmental sequence relies on longitudinal studies of stability and growth for

people over time^{36–39} and cannot be evaluated in the present data. Several features of our data are nonetheless relevant to the structural claim that ego stages describe an ordered, developmental sequence and not merely a set of unordered categories. One line of evidence relevant to the ordinal character of ego development is that adjacent stages should be more similar than progressively more distant ones. In other words, correlations between stages should adhere to a simplex model⁴⁰.

In Supplementary Table 2, correlations between ego levels in word use are presented both for the total sample of 10,670 words (above the diagonal) and the curated sample of 1,811 words which occur 10 or more times in the combined sample (below the diagonal). In both cases, the expected ordering is satisfied for 29 of the 30 pairwise comparisons; that is, with one exception, correlations become progressively smaller as one moves vertically or horizontally away from the diagonal. In both the total and the curated samples of words, this exception (represented in italics in Supplementary Table 2) occurs at the Conformist stage, which appears less similar to the previous Self-protective stage than predicted by a simplex model. When the empirical and exemplar data are analysed separately, this one reversal is present only in the former; the exemplar data show a perfect simplex. This result supports both the ordinal character of stages of ego development and the present method of assessing ego level through word counts (for code, see CorrsBetLevels0218.Rmd at <https://osf.io/jw7dy/>).

(2) Ego level may be seen as an ordered sequence of milestones that subsume or build on previous milestones, that is, as an additive-stage model⁴¹. This has implications for the cognitive facet of ego development: with higher stages, responses to items on the WUSCT should become longer and, arguably, more differentiated⁴². As in previous studies, we found an association between verbosity and ego level^{43,44}. Furthermore, the expected number of distinct words in a single response also increased with ego level. These trends held for each of the four constituent samples individually and across all ego levels, with only one exception—in the sample of midlife adults, responses at the Conformist stage were shorter than those at the previous stage; see FluenciesbySample0218.Rmd at <https://osf.io/jw7dy/>. As ego develops, responses become longer and vocabulary becomes richer (see Supplementary Table 1).

In addition to simple counts of words, deeper characteristics of speech should also show increasing complexity over the course of development. In previous studies with the LIWC, cognitive complexity has been assessed in various ways: in one study⁴⁵, complexity was measured from a combination of ten LIWC indicators (such as causation and insight), in another⁴⁶, a simpler combination of four indices was used (including differ and negate). Still another approach has been to measure complexity solely from function words⁴⁷. We standardized each of these three measures and combined them into a single index of LIWC cognitive complexity and

then examined the relationship between this composite measure and ego level. With the exception of the highest and lowest stages, confidence intervals for adjacent ego levels were non-overlapping. The effect size for the difference in complexity between the highest and lowest stages was 0.72, and was at least 0.64 in each of the four individual samples. Even among the most frequently encountered stages in the middle ranges of ego development, differences in complexity were non-trivial, with a difference between the Conformist and the Conscientious stages of $d=0.41$ and at least 0.35 in each of the four individual samples (Fig. 1a). Responses to the WUSCT become not only longer but also more complex with progressively higher stages of ego development (see LIWCComplexity0218.Rmd at <https://osf.io/jw7dy/>).

(3) A core concept in social-cognitive development is that of perspective taking, which develops from egocentrism, through an ability to infer the views of others, to a detached third-party perspective, to, at the highest stages, a perspective that considers the system as a whole⁴⁸. This progressive broadening of perspective suggests that one might anticipate relationships between pronoun usage and ego development, with preferences for first-person singular, first-person plural, second-person, third-person and impersonal stances describing an ordered sequence. However, previous studies of pronoun usage with the LIWC have suggested that the relationship between pronouns and personality is nuanced. For example, although researchers have found an increase in the use of first-person plural pronouns (we) following the cultural trauma of 9/11 (ref. ⁴⁹) and have argued that the use of third-person pronouns (he, she) suggests a focus on others instead of the self⁵⁰, a first-person singular stance does not appear to be associated with narcissism⁵¹, but instead with depression⁵² and interpersonal distress⁵³.

We examined LIWC categories for first-person singular (i), first-person plural (we), second-person (you), third-person (combining she/he and they) and impersonal (ipron) pronouns. Relationships between these forms of pronoun use and ego level are illustrated in the remaining panels of Fig. 1. The use of 'I' was particularly characteristic of the Self-protective stage (mean $z=0.27$) and declined at higher stages (Fig. 1b). Similarly, impersonal pronouns showed the expected trend of increased prevalence at higher ego levels (Fig. 1f).

However, there was no evidence that the use of first-person plural or second-person pronouns were associated with intermediate levels, nor were third-person pronouns associated with high ego levels (Fig. 1c–e). With development, pronoun use changes away from 'I' and towards 'its', but other expected trends in pronoun use were not evident (see Multiplot0218.Rmd at <https://osf.io/jw7dy/>).

(4) Additional relationships between LIWC categories and ego development were explored. To the extent that ego development is best understood as a series of achievements rather than as a single polar variable, relationships between development and other characteristics should deviate from a simple linear trend. To examine the nature, magnitude and robustness of the relationships between the LIWC and ego level, we used k-fold cross-validation. For each LIWC category, the total set of 43,993 WUSCT responses was divided into 10 roughly equal samples, each of which was used to test the results of polynomial regression analyses undertaken on the remaining 90% of the data; the results of these cross-validated analyses were then averaged. For 27 of the 81 LIWC categories, a cubic model showed the best fit, for 19, a quadratic model, and for the remaining 35 LIWC categories, the best-fitting model was linear. Then, to further examine the robustness of these linear and non-linear trends, we used the same method of k-fold cross-validation on each of the constituent samples. Key results are shown in Table 2. Summary statistics for LIWC categories by ego level are included in the Supplementary Data.

In the combined data, 26 LIWC categories showed non-zero ($R^2 \geq 0.01$) cross-validated relationships with ego level. For some LIWC themes, relationships with ego level were essentially linear, including those for cognitive processes, complexity and achievement (which increase over ego development), and verbs and some perceptual processes (which show a decline). However, for 18 LIWC categories, the best-fitting model included modest quadratic or cubic effects. Although R^2 increments were invariably small (median = 0.003 in the combined sample), their shape or direction was found to be largely consistent across the constituent datasets. LIWC categories associated with informality (informal, leisure, assent) and impulse (anger, swear words, sexual, body, ingestion) are associated with declining ego level, but with some levelling-off

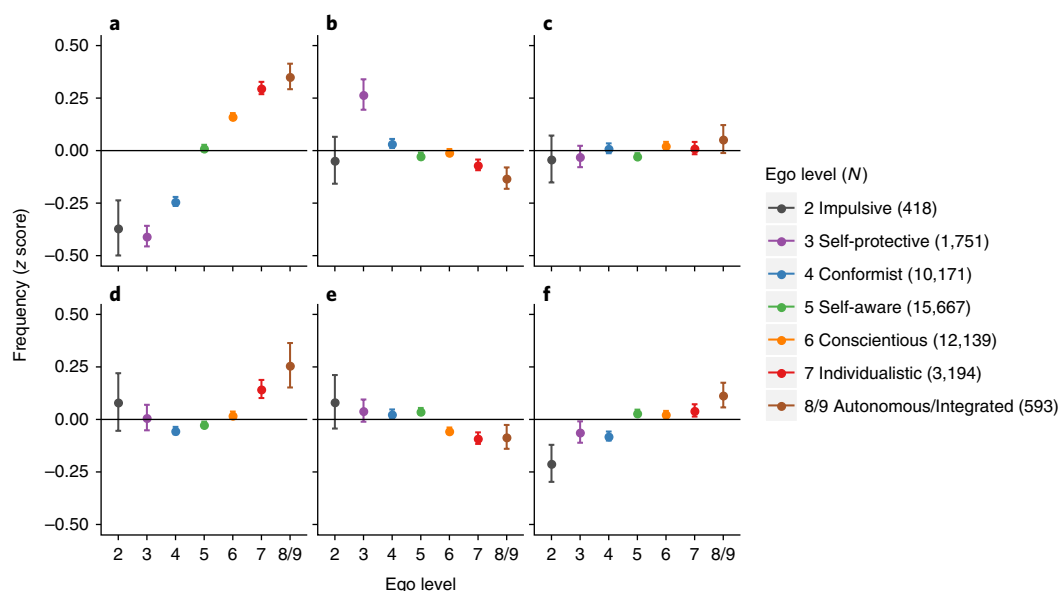


Fig. 1 | Ego level and the LIWC. With development, complexity increases, but pronoun use indicates mixed support for increasing breadth of perspective. **a**, Complexity (average of three measures^{45–47}). **b**, First-person singular pronoun (i). **c**, First-person plural (we). **d**, Second-person (you). **e**, Third-person singular and plural (she/he, they). **f**, Impersonal (ipron). *N* refers to the number of coded responses at each ego level. Error bars represent means ± 1.96 standard error. See Multiplot0218.Rmd at <https://osf.io/jw7dy/>.

Table 2 | LIWC categories as a function of ego level: coefficients in combined data and trends in individual samples

Trend and LIWC theme	Combined data						Constituent samples				
	Model fit		Coefficients				Shape of best model (sparkline)				
	Linear r	R^2	Intercept	Linear	Quadratic	Cubic	All	College (L.A.K.)	College (K.L./R.E.P.)	Midlife (D.P.M.)	Exemplar
Increasing and concave upward trends for verbosity											
Words per sentence	0.45	0.21	4.87	-1.25	0.31						
Word count	0.40	0.18	4.38	-0.51	0.05	0.03					
Increasing and s-shaped trends for some function words											
Conjunctions	0.24	0.06	11.93	-8.38	2.00	-0.13					
Prepositions	0.10	0.01	5.35	0.47	0.24	-0.02					
Articles	0.07	0.01	8.55	-3.97	0.94	-0.06					
Increasing trends for cognitive processes, complexity and achievement											
Differentiation	0.13	0.02	6.47	-4.14	1.01	-0.06					
Cognitive processes	0.11	0.01	0.79	3.11	0.04	-0.02					
Words >6 letters	0.09	0.01	0.56	5.95		-0.42					
Insight	0.10	0.01	-0.71	0.74							
Comparatives	0.10	0.01	-0.91	0.63							
Achievement	0.08	0.01	-0.08	0.55							
Concave downward trends for emotional tone and social processes											
Emotional tone	0.09	0.01	23.48	6.16	0.00	-0.04					
Social processes	-0.06	0.01	9.90	4.46	-0.55						
Decreasing and concave-upward trends for informality and impulse											
Informal speech	-0.05	0.01	8.75	-3.95	0.60	-0.03					
Swear words	-0.05	0.01	4.25	-2.18	0.37	-0.02					
Assent	-0.05	0.01	6.38	-3.34	0.58	-0.03					
Anger	-0.05	0.01	10.32	-4.87	0.80	-0.04					
Sexual	-0.06	0.02	11.27	-5.92	1.02	-0.06					
Ingestion	-0.07	0.01	6.16	-2.72	0.40	-0.02					
Body	-0.09	0.01	5.64	-1.67	0.13						
Leisure	-0.07	0.01	4.62	-0.99	0.06						
Present-focus	-0.16	0.03	30.63	-2.28							
Decreasing trends for verbs and some perceptual processes											
Perceptual processes	-0.09	0.01	8.93	-1.01							
Hear	-0.09	0.01	6.34	-0.90							
Auxiliary verbs	-0.11	0.01	20.71	-1.40							
Common verbs	-0.17	0.03	36.50	-2.70							

Points in sparklines represent mean LIWC category scores as a function of ego level (see Table 1). Sparklines illustrate the shape of relationships. Shading indicates the degree of best-fitting models following 10-fold cross-validation: grey, linear; yellow, quadratic; and orange, cubic. Only LIWC categories associated with ego level at $R^2 \geq 0.01$ are shown. Abbreviations (for example, L.A.K.) refer to the authors' laboratories from which the data were drawn. N : all = 43,933, college (L.A.K.) = 4,364, college (K.L./R.E.P.) = 26,558, midlife (D.P.M.) = 5,418 and exemplar = 7,593.

following the first few stages of development. Function words (conjunctions, prepositions and articles) show a pattern of accelerated use through the middle stages, whereas the relationship of ego development to verbosity or response length appears concave upwards, with greater effects at the later stages of development. This contrasts with the pattern for social processes and emotional tone, which show trends of levelling or decline at the highest stages. Taken together, these trends provide support for the structural claim that ego level can be understood as a set of qualitatively distinct stages as well as a single dimension (LIWCRegressions0218bySample.Rmd at <https://osf.io/jw7dy/>).

(5) To represent relations between ego stages and individual words, the characteristic language for each ego level is summarized in a comparison word cloud (Fig. 2), in which the 250 most distinguishing words are placed according to polar coordinates that correspond to ego level (90- Φ) and characteristic strength ($-r$). Ego level progresses clockwise, beginning with the lowest (Impulsive) level (in black, at the top of Fig. 2), through intermediate levels, including Conformist (red) and Conscientious (green), to the combined

Autonomous/Integrated stages (purple) in the upper left of Fig. 2. The most characteristic terms are large and close to the origin (ComparisonCloud0218.Rmd at <https://osf.io/jw7dy/>).

Few responses (<1% of the empirical responses and <2% of exemplars) are classified at the lowest (Impulsive) stage of development; furthermore, language at this level is, in comparison with more-advanced stages, relatively impoverished. Words that were most characteristic of this stage in both the comprehensive and the stratified datasets included terms related to sex (ass, blow, laid, sex, sucks), aggression (bothers, fight, hate, violence) and the consumption of food and drink (eat, meal, drink, dinner), and are consistently with an interpretation of the Impulsive stage as being largely concerned with primary biological drives. Language reflecting the secondary motive of acquisitiveness was also in evidence (buy, gets, receives, shopping), as was a cluster of simple, positively toned evaluative terms (amazing, awesome, cool, fine, OK, okay, nice).

The Self-protective stage is still relatively rare in our data, occurring in <4% of responses. The largest cluster of terms was associated with interpersonal attraction (beautiful, dance, eyes, hair,

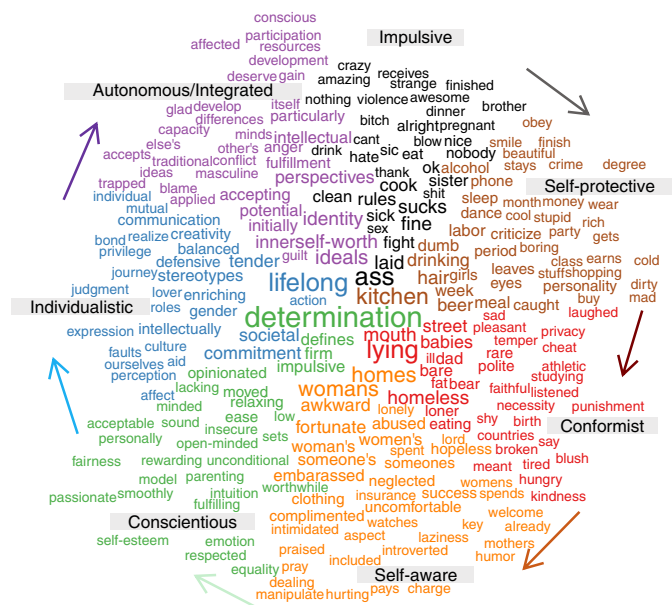


Fig. 2 | A comparison word cloud showing the 250 terms showing the strongest association with ego level. The most distinguishing words for each of seven ego levels are shown. Ego level progresses clockwise, beginning with the lowest (Impulsive) level (in black, at the top of the figure), through intermediate levels, including Conformist (red) and Conscientious (green), to the combined Autonomous/Integrated stages (purple) (see ComparisonCloud0218.Rmd at <https://osf.io/jw7dy/>).

personality, smile wear); one interpretation is that the focus, unlike the previous stage, is on an appetitive rather than a consummatory phase of drive reduction. Other terms were associated with alcohol consumption (alcohol, beer, drink, drinking) and food preparation (cook, kitchen, meal).

The Conformist stage comprises 25% of the empirical responses. Although some of the characteristic terms suggest social charm (listened, pleasant, polite) or virtue (faithful), a larger number suggests struggle (blush, meant, say, shy, temper) and falling short (broken, cheat, lying). Among other characteristic terms are nouns evocative of banal positive (animals, babies) and negative affect (homeless, ill, loner, sad, sick) and, as at previous stages, food (eat, eating, fat, mouth).

The Self-aware stage is the modal level in our data, including 37% of the empirical responses. The characteristic language appears increasingly differentiated and includes a focus on the self (I'm, introverted), and, with it, indications of affective experience associated with being in the spotlight⁵⁴ for better (complimented, praised) or worse (awkward, embarrassed, nervous, uncomfortable), consistent with a nascent and socially contingent self-esteem⁵⁵. The focus on impulse, which characterized the earliest stages, is supplanted by distal concerns with issues ranging from faith (lord, pray) and fortune (fortunate, abused, neglected) to financial responsibility (charge, clothing, homes, pays, spends, spent, insurance) and gender roles (woman's, women's).

The Conscientious stage comprises 27% of the empirical responses, with characteristic language largely focused on facets of achievement motivation, including drive (contribute, determination, firm, opinionated, passionate), relaxation (ease, relaxing) and the evaluation of outcomes (acceptable, evaluate, fulfilling, rewarding, worthwhile). In addition, there are terms describing psychological characteristics, including (self-)esteem, impulsive, insecure, (open-)minded, parenting and unconditional.

Unlike the preceding three stages, the Individualistic stage is rare, appearing in only 6% of our empirical responses (but in 15% of the exemplars). Language is suggestive of intellectual growth (creative, creativity, culture, enriching, fulfillment, intellectual,

intellectually) and of mutuality in interpersonal relationships (aid, balanced, bond, commitment, communication, lover, mutual, ourselves, tender). There is a deeper conception of the self (gender, identity, individual, perception, privilege, stereotypes), its limitations (defensive, faults, harm) and its development (finally, initially, lifelong, lifetime, realize, throughout).

Finally, responses at the Autonomous and Integrated stages occur in <1% of our empirical responses and are characterized by a continuing emphasis on the self (identity, ideals, inner, masculine, stereotypes) and language which suggests compromise or the renegotiation of life goals (accepting, perspectives, gain, sacrifice).

Considered together, these analyses of approximately 44,000 sentence completions provide a range of evidence relevant to the construct validity of Loevinger's¹ model. Relations between language and ego level were largely consistent across different sets of items. Responses at higher ego levels were not only more verbose but were also characterized by more-complex speech. The frequency of word use at different stages adhered to a simplex model, supporting the ordinal sequence of stages. Moreover, the language associated with each stage was distinct and coherent. As shown in Table 2, the course of ego development tracked polynomial trends for several LIWC categories: anger, ingestion and swear words were particularly associated with the lowest stages, social processes and articles with intermediate stages, and cognitive processes and achievement with the highest stages of ego development. The specific terms associated with particular stages and transitions mirrored an apparent shift from consummatory to appetitive desires at the lowest stages, the dawn of skepticism regarding the self and the world at the Self-aware stage and the centrality of achievement motivation at the Conscientious stage. These findings contribute to a deeper understanding of the construct of ego development and its relationship with other models of personality and individual differences^{56–58}.

The linguistic results reported here reflect at least three developmental trends or currents that run within the broad stream of what Loevinger called ego development. The first is a trend towards complexity. Letters per word and the number of words per response both increased with ego development scores, as did individual LIWC categories that were indicative of complexity, such as conjunctions, comparatives and differentiations. Second is a move from egocentrism to a broader social referent, for which the results provided mixed support. In support of this conjecture, the relative frequency of first-person singular pronouns declined with development, as did language focusing on the present, whereas the use of more-detached, impersonal pronouns increased. However, second-person and third-person pronouns did not show the concave quadratic trends that might be expected with Loevinger's proposed sequence.

The third trend is the waxing and waning of constraint. The conflict between the self's agency on the one hand and the limitations placed on the self by the social group on the other is probably as old as human nature itself. Loevinger's theory captures the changing features of this dynamic as the person moves from lower to higher stages of ego development. At the earliest (Impulsive) stage in our data, language is essentially solipsistic. At the subsequent Self-protective stage, terms such as 'caught' and 'obey' appear, then, at the Conformist stage, 'lying' and 'cheat', indicating a growing awareness that others exist and that norm violations may bring explicit sanctions. Constraints seem to become more internalized at the next stage (Self-aware), as indexed in references to self-conscious emotions such as embarrassment and shame. Here, terms such as 'awkward', 'embarrassed' and 'uncomfortable' begin to appear. Higher stages suggest greater internalization still and a dampening of concern regarding sanctions for norm violation. Instead, the person may become more concerned with conflicts regarding achievement (Conscientious stage), mutuality (Individualistic stage) and, ultimately, self-actualization (Autonomous/Integrated stages). Rather than worrying so much about committing misbehaviour, the

person becomes more concerned with living up to his or her own standards regarding what a mature human being—who is both a motivated agent and a member of society—should do.

Although the kind of linguistic methodology put forth in this article may ultimately promote more research on ego development, an important conceptual question for personality psychology still remains: where does ego development fit within the broad gamut of constructs that comprise personality? If one limits personality to basic dispositional traits, such as those captured in the Big Five, ego development would appear not to fit in very easily. Dispositional traits, such as extraversion and conscientiousness, capture basic individual differences in socio-emotional style and performance. A person with high extraversion tends to exhibit an enthusiastic, gregarious, socially dominant, reward-seeking style of social performance across a wide range of situations and contexts, compared (on average) to a person with low extraversion. However, the linear dimension, running from low to high, does not entail an explicit developmental logic or structure, unlike Loevinger's stage model for ego development. There is no expectation, for example, that a person will naturally develop from low extraversion to high extraversion, increment by increment, over time. Even as certain traits, such as conscientiousness and agreeableness, exhibit normative increases from adolescence into middle adulthood², they do not in and of themselves specify a stage-like, developmental logic, nor does openness to experience, which is the one Big Five trait with which ego development shows consistently positive associations¹⁵.

Contemporary models of personality that go beyond traits may open up more conceptual space for ego development. In this regard, it has been argued^{59,60} that personality comprises three layers: dispositional traits (the person as a social actor), characteristic adaptations (goals and values mainly; the person as a motivated agent) and integrative life narratives (the person as an autobiographical author). Traits track socio-emotional behavioural trends across current contexts, goals and values capture the person's motivational agenda for the future, and life stories (narrative identity⁶¹) convey how a person makes narrative meaning out of the remembered past, experienced present and the anticipated future. Within this framework, ego development would appear to run through the second and third layers of personality, structuring motivation and meaning. Indeed, research has shown that higher levels of ego development are associated with (1) more value questioning and more humanistic value profiles⁶² and (2) greater levels of plot complexity and character subtlety in life stories^{63,64}. Research also indicates that narrators who are higher in ego development tend to tell life stories that emphasize themes of growth and development⁶⁵.

By way of limitations, the current study focuses exclusively on language conveyed through sentence completions. Other verbal forms imply different demand characteristics and call for different speech content. The present data are limited too, by the relative homogeneity of the research participants. Over 70% of the responses were drawn from relatively well-educated, white samples of American and Australian university students. Moreover, the linguistic methods used to infer ego level from text are, from the standpoint of computational linguistics, quite primitive. Alternative approaches might include, for example, predicting ego level from multi-word strings or *n*-grams⁶⁶, aggregating responses over variant spellings (embarrassed, embarrassed) or common stems (creative, creativity), as well as more ambitious attempts to disambiguate words or to extract the meanings of words (bacon, sausage) from their covariation with other terms in various large corpora^{67,68}. Alternative approaches such as these, investigated in richer and larger datasets, may lead to the discovery of more robust and comprehensive links between personality and language.

Although the present study sheds substantial light on the construct of ego development and its constituent stages, it only scratches the surface of how the methods of data science can inform

the study of personality development. We have not yet examined the question of the extent to which ego level can be assessed computationally from the WUSCT, nor have we attempted to empirically resolve the related question of how many stages can be reliably distinguished from samples of text. We have examined only sentence completions and not richer narrative accounts of the life history; at least one investigation has found that, in such accounts, codes by human raters (for redemption sequences) and those by automated analysis (LIWC categories) are essentially complementary⁶⁹. In addition, our sample, although large from the standpoint of traditional personality measurement, is nonetheless too small to uncover stable relationships between less common words, such as 'cycle' and 'phonies', and ego level, particularly at the highest and lowest stages of development.

Nonetheless, although we are agnostic regarding the extent to which the automated assessment of ego level could be used as a tool for making decisions about individuals, we do believe that the methods show promise as instruments for assessing cultural and temporal progress (or its absence) in written texts, including on social media platforms. Culturomics, or the assessment of culture through samples of language, is challenging^{70,71}. Nonetheless, if ego development can be scored from everyday language, then the content of texts from Twitter feeds to political speeches, and from children's stories to strategic plans, may provide new insights into our state of moral, social and cognitive development.

Methods

Data include empirical responses collected between 1989 and 2014 in the laboratories of three of the authors (L.A.K., K.L. and D.P.M., with 4,364, 26,558 and 5,418 sentence completions, respectively) and exemplars (7,593 responses scanned from the 1996 manual¹⁰). All participants completed and signed informed consent forms. All data reported in this manuscript were collected as part of institutional review board-approved protocols at the institutions where data were collected. All federal ethical guidelines have been followed during the collection and reporting of these data.

The first two empirical samples (L.A.K. and K.L.) were students at four universities in Australia and the United States; these data were typically scored by two or more judges (79% or 24,351 cases) and include 6,156 responses assessed by the remaining author (R.E.P.) as part of her undergraduate honors thesis⁷². The remaining sample (D.P.M.) was composed of midlife (55–58 years of age) adults, approximately 45% African American and 55% white, from the Foley Longitudinal Study of Adulthood and were scored by a single trained judge. All but 819 responses were completions of one of the 18 stems in Form 81 of the WUSCT¹⁰; the remaining responses were from the 36-item version of the instrument. For data collected before 1995, ego level was recoded using the scoring key provided in the 1996 manual¹⁰. Where two judges rated ego level and the average of these ratings was midway between adjacent stages, the higher level was used to provide an integer score for each response. In published studies of subsets of these data, the ratio of female to male respondents is approximately 2/1 and interjudge reliabilities ranged from 0.87 to 0.94 (refs ^{73–75}).

Loevinger's theory of ego development posits nine stages, the earliest of which is preverbal and so does not appear in responses to the WUSCT (see Table 1). Of the remaining eight stages, the highest is quite rare, appearing in only 14 responses in our datasets. Accordingly, the highest two stages were combined into an autonomous/integrated stage, which is now represented by 593 responses. Although others have argued for an extension of Loevinger's model to a still-higher set of ten stages of ego development⁷⁶, no more than seven distinct stages are discernible in responses to the WUSCT. Across these stages, the number of responses ranged from 418 at the lowest (Impulsive) stage to 15,667 for the modal Self-aware stage (see Supplementary Table 1).

Effects of data source and sentence stem. Sentence completions were drawn from four sources, including three empirical datasets and the exemplars or scoring manual. Differences between the empirical datasets in ego level were modest. With ego level scored as a continuous dimension, mean values ranged from 4.91 to 5.25 and standard deviations (s.d.s) from 1.01 to 1.03. Collectively, the average ego level across the responses in the empirical data was 5.05, corresponding to the Self-aware stage, and similar both to that reported in a national probability sample of adults⁷⁷ and to the average score for college and young-adult samples in a previous meta-analysis³⁶. In the exemplar data, which included all of the scored responses in a previously published manual¹⁰, the average rating was slightly higher (5.45), resembling the mean for adult samples in a meta-analysis³⁶, and slightly more variable (s.d. = 1.27). The higher mean and greater variability in the exemplar data reflect a disproportionate number of responses at extreme levels,

particularly at the high end, and reflects the origin of this dataset as a guide for scoring rather than a representative sample of responses to the instrument (from EgoLevelBySample0218.Rmd at <https://osf.io/jw7dy/>).

Comparisons of groups of sentence stems. If the present attempt to assess ego level from natural language is to be successful, the relationship between language and ego level cannot be substantially dependent on particular WUSCT stems (for example, item 4: "A man's job..." versus item 18: "Rules are..."). To explore this, we split the core items in Form 81 into three sets of six stems each, then computed word frequencies for each ego level for each of these. Correlations between these samples of word counts provide a measure of the extent to which word use at each level is robust across sets of stems or stimuli. Across samples of stems and ego levels, the median alpha coefficient summarizing these correlations was 0.79. There was less convergence at the lowest (Impulsive) stage, in part because responses at this level are relatively rare and the speech relatively impoverished ($\alpha=0.37$). At the remaining stages, alpha coefficients ranged from 0.68 to 0.89. Similar results are obtained if responses to items 19–36 are included as well. This analysis indicates that the terms that characterize individual ego levels are largely, although not entirely, independent of specific SCT stems and provides justification for a closer examination of the relationship between words or word categories and ego level (AlphaForItems0218.Rmd at <https://osf.io/jw7dy/>).

Cleaning and reduction of text. Using the R package Quanteda⁷⁸, data were cleaned in a four-step process. First, all responses were expressed as lower-case and selected data features were removed, including most punctuation (all but hyphens and apostrophes). The set of sentence completions was, at this point, represented as a single matrix of 10,670 words $\times$ seven ego levels in which entries correspond to the count of occurrences of each word at each level. Second, we removed words that appeared in response to only one or two stems, as these were judged to be less likely to characterize ego level in forms of speech outside of sentence completions. Third, words were removed that appeared <10 times in the total sample. (Although relatively unusual words, such as racially and exercise, might be expected to be associated with ego level, such relations are inherently unreliable and may be spurious.) Finally, we removed all non-words consisting of a single character (for example, 'b'), as we judged that these were most likely in the data owing to transcription or scanning errors. This led to a set of 1,811 terms, most of which (1,593 or 88%) were at least linearly related to ego level ($P < 0.05$, corrected for false discovery rate; see Clean0218.Rmd at <https://osf.io/jw7dy/>).

Assessing ego level as an ordered set of categories. To assess the characteristic language associated with each stage or level of development, we computed weighted risk ratios for each word and stage. That is, because ego development is modelled as an ordered set of categories, we divided the proportion of the occurrences of each word at each stage by the proportion of that word for each of the remaining levels, with these latter proportions weighted by their distance from the target stage. For example, to assess the relevance of the word 'happy' for the Conformist stage, each occurrence at the (adjacent) Self-aware stage would be weighted -1 , whereas those at the more distant Conscientious stage would be weighted -2 , and so on. These values were then standardized by column to facilitate comparisons between ego levels. This analysis was undertaken on both the combined data and the stratified (but necessarily incomplete) dataset in which the proportion of data from each sample was held consistent across ego levels. Complete results for all 1,811 terms for both analyses, as well as the individual samples, are provided in Supplementary Data.

Life Sciences Reporting Summary. Further information on experimental design is available in the Nature Research Reporting Summary linked to this article.

Code availability. R markdown (Rmd) files for all analyses presented in this paper can be found at the Open Science Framework repository <https://osf.io/jw7dy/>.

Data availability The authors received no specific funding for this work. The raw data (a spreadsheet including the sample, item number, sentence completion, rating and number of raters) are available at the Open Science Framework repository <https://osf.io/9sbhw/>.

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Author contributions

K.L. designed the research and analysed the data. All authors collected the data and contributed to the writing of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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► Experimental design

1. Sample size

Describe how sample size was determined.

The sample comprises 43,933 responses. The study is a reanalysis of existing data, this is the complete set of scored sentence completions that were available to us.

2. Data exclusions

Describe any data exclusions.

No data were excluded from the analysis.

3. Replication

Describe the measures taken to verify the reproducibility of the experimental findings.

Reproducible results (on the present set of data) are assured by the availability of data and code at the OSF repository. The replicability of the present results is made more likely by (a) the large sample size, (b) the use of techniques explicitly addressed to this end, including k-fold cross-validation.

4. Randomization

Describe how samples/organisms/participants were allocated into experimental groups.

n.a., please see above.

5. Blinding

Describe whether the investigators were blinded to group allocation during data collection and/or analysis.

n.a.

Note: all in vivo studies must report how sample size was determined and whether blinding and randomization were used.

6. Statistical parameters

For all figures and tables that use statistical methods, confirm that the following items are present in relevant figure legends (or in the Methods section if additional space is needed).

n/a Confirmed

- ☐ ☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement (animals, litters, cultures, etc.)
- ☐ ☒ A description of how samples were collected, noting whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☒ ☐ A statement indicating how many times each experiment was replicated
- ☐ ☒ The statistical test(s) used and whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐ ☒ A description of any assumptions or corrections, such as an adjustment for multiple comparisons
- ☐ ☒ Test values indicating whether an effect is present
*Provide confidence intervals or give results of significance tests (e.g. *P* values) as exact values whenever appropriate and with effect sizes noted.*
- ☐ ☒ A clear description of statistics including central tendency (e.g. median, mean) and variation (e.g. standard deviation, interquartile range)
- ☐ ☒ Clearly defined error bars in all relevant figure captions (with explicit mention of central tendency and variation)

See the web collection on [statistics for biologists](#) for further resources and guidance.

► Software

Policy information about [availability of computer code](#)

7. Software

Describe the software used to analyze the data in this study.

R

For manuscripts utilizing custom algorithms or software that are central to the paper but not yet described in the published literature, software must be made available to editors and reviewers upon request. We strongly encourage code deposition in a community repository (e.g. GitHub). *Nature Methods* [guidance for providing algorithms and software for publication](#) provides further information on this topic.

► Materials and reagents

Policy information about [availability of materials](#)

8. Materials availability

Indicate whether there are restrictions on availability of unique materials or if these materials are only available for distribution by a third party.

no restrictions

9. Antibodies

Describe the antibodies used and how they were validated for use in the system under study (i.e. assay and species).

n.a.

10. Eukaryotic cell lines

a. State the source of each eukaryotic cell line used.

n.a.

b. Describe the method of cell line authentication used.

n.a.

c. Report whether the cell lines were tested for mycoplasma contamination.

n.a.

d. If any of the cell lines used are listed in the database of commonly misidentified cell lines maintained by [ICLAC](#), provide a scientific rationale for their use.

n.a.

► Animals and human research participants

Policy information about [studies involving animals](#); when reporting animal research, follow the [ARRIVE guidelines](#)

11. Description of research animals

Provide all relevant details on animals and/or animal-derived materials used in the study.

n.a.

Policy information about [studies involving human research participants](#)

12. Description of human research participants

Describe the covariate-relevant population characteristics of the human research participants.

See Table S1 for sample characteristics. 70% (30922) of the responses are drawn from college student samples, 12% (5418) from midlife adults, and 7593 (17%) from the scoring manual for the test. Analyses of the robustness of results across these samples (i.e., word use by ego level in each sample) is included in Table S3.